**BACHELOR OF
ENGINEERING HONORS DEGREE IN
TELECOMMUNICATIONS ENGINEERING**

Integrated Early Detection of Contagious Broiler Diseases system

PROJECT BY:

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2) CHAPTER TWO: LITERATURE REVIEW

2.1 INTRODUCTION

This chapter provides a review of existing literature related to the early detection of diseases in poultry farming, with a specific focus on applications of machine learning, deep learning, and edge computing. It examines existing systems currently utilized in the detection of Coccidiosis and Newcastle Disease. The chapter further analyzes existing AI-based poultry disease detection systems, identifying and highlighting key research gaps. Additionally, it compares various deep learning models employed for image-based disease detection, including YOLO, FOMO, MobileNetV2, and other object detection/classification models.

2.2 REVIEW OF OTHER RELEVANT SYSTEMS

This section explores existing systems or approaches used for disease detection in broiler farming.

2.2.1 Sensor-Based Systems

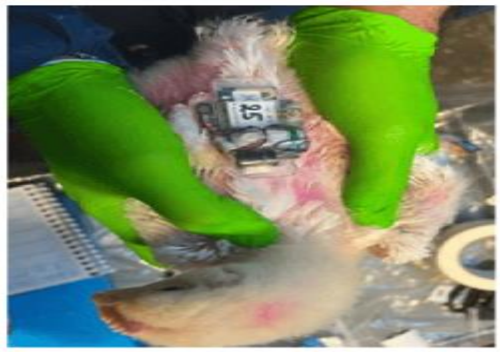
2.2.1.1 Wearable Sensors

Wearable sensor systems represent an innovative approach to broiler health monitoring, providing real-time insights into bird behavior, activity, and potential signs of disease. These systems utilize Inertial Measurement Units (IMUs) and Ultra-Wideband (UWB) sensors to track movement, detect stress levels, and identify early symptoms of illness in broiler chickens.

One existing system integrates IMUs and UWB sensors to continuously monitor acceleration, movement patterns, and spatial positioning of individual birds. This system was designed to address challenges in large-scale commercial poultry farming by enabling farmers to detect abnormal behavior patterns indicative of disease or stress. The research evaluated different methods for attaching sensors to the birds, concluding that suturing provided the most effective long-term retention. The IMU sensors were successful in differentiating individual activity variations and environmental influences, while the UWB sensors tracked up to 200 birds simultaneously with high spatial accuracy. However, accuracy was affected by sensor placement and farm layout, which required careful calibration.[5]

This wearable sensor system offers a scalable and data-driven approach to poultry health monitoring, enhancing early disease detection, improving ethical farming practices, and increasing overall farm productivity. By integrating such automated sensor-based disease detection systems, poultry farmers can reduce mortality rates and optimize farm management.

The image below shows the wearable sensor being used.



2.2.1.2 Environmental Sensors

The poultry industry is increasingly adopting sensor-based systems to monitor and control environmental factors that significantly impact broiler health and productivity. A primary focus of these systems is the management of stress, as environmental stressors can have profound negative effects on poultry. As highlighted in [1], birds are homoeothermic and maintain a stable internal temperature within a thermoneutral zone. When environmental temperatures deviate from this zone, broilers experience heat or cold stress, leading to reduced feed intake, decreased growth, impaired immune function, and increased susceptibility to disease[1].

Environmental monitoring systems utilize various sensors to measure key parameters within the poultry house. Temperature and humidity sensors are critical for maintaining optimal thermal conditions and preventing heat and cold stress. As birds are highly sensitive to heat stress due to the absence of sweat glands and their feather covering[1], precise temperature control is essential, particularly for fast-growing commercial broilers that are more susceptible to heat-related stress[1]. Additionally, sensors can monitor air quality parameters such as ammonia levels, which, when elevated, can cause respiratory distress and compromise bird health. By providing continuous and real-time data on these environmental factors, sensor-based systems enable farmers to make informed decisions and implement timely interventions to mitigate stress and optimize broiler well-being.

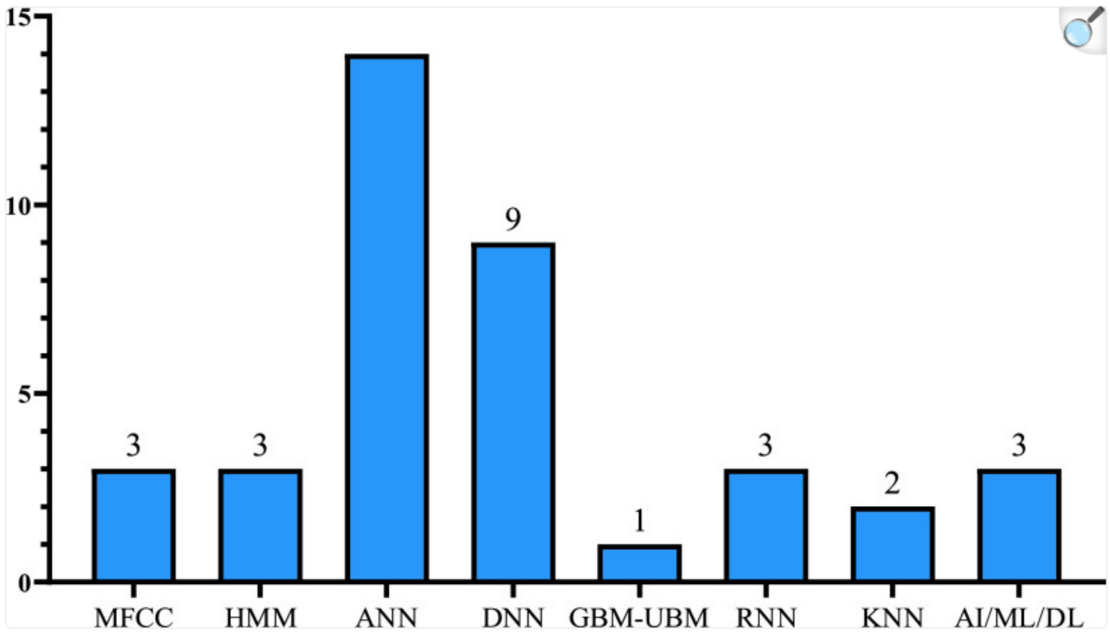
2.2.2 Sound Analysis Systems

Sound analysis systems represent a promising avenue for the early detection of respiratory diseases in poultry. The development of these systems is predicated on the fact that respiratory diseases often manifest in altered vocalizations, such as coughing, sneezing, and abnormal respiratory sounds[7]. Microphones are employed to capture audio within poultry houses, and signal processing techniques are used to analyze the recorded sounds for patterns indicative of disease.

The application of sound-based technologies is particularly relevant for broiler production due to the potential for non-invasive and continuous monitoring of flock health. As highlighted in [7], a systematic review of precision livestock farming (PLF) technologies found that sound-based technologies were the most common approach for monitoring respiratory diseases in swine. While the review included fewer poultry studies (6), it emphasizes the potential of sound analysis for detecting respiratory ailments across different livestock species

However, the review also indicates that the reliability of sound-based systems for automatic detection of respiratory diseases in livestock remains a challenge. As stated in [7], many studies were conducted under laboratory conditions rather than field conditions, and some used unreliable reference tests. This highlights the importance of rigorous validation in real-world settings to ensure the accuracy and effectiveness of sound-based disease detection systems. The complexity of diagnosing respiratory diseases, which often involves assessing multiple clinical signs beyond just coughing, further underscores the need for sophisticated sound analysis techniques and potentially the integration of sound data with other monitoring modalities. This multimodal approach is a key component of the proposed system, which combines sound analysis with image analysis for improved disease detection

Below is a picture showing popular AI/ML techniques used in the current cough-based detection and diagnosis approaches.



Several key technologies are utilized in the development of sound-based cough detection systems. Mel-Frequency Cepstral Coefficients (MFCCs) are frequently employed for feature extraction, representing the spectral envelope of audio signals. Hidden Markov Models (HMMs) are used to model the temporal evolution of cough sounds, capturing the dynamic changes in frequency and amplitude over time. Machine learning algorithms play a crucial role in classification, including Logistic Regression (LR), Artificial Neural Networks (ANNs), Support Vector Machines (SVMs), Random Forests (RF), Deep Neural Networks (DNNs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs). Principal Component Analysis (PCA) is utilized for dimensionality reduction, simplifying feature sets while preserving essential information. Wavelet transforms (CWT) are used for time-frequency analysis, allowing for the identification of transient sound events like crackles. Additionally, platforms like TensorFlow and devices like Raspberry Pi are used for implementation and deployment of these systems, including mobile applications that leverage smartphone capabilities. Techniques like Gaussian Mixture Models (GMMs) and k-Nearest Neighbors (k-NN) are also used for classification, and Long Short-Term Memory (LSTM) networks are used for sequential data analysis, particularly in COVID-19 cough detection.[13]

2.2.3 Computer Vision-Based Systems

The poultry industry faces challenges such as workforce shortages and incomplete data collection, hindering uniformity and traceability. To address these issues, an intelligent video surveillance system was implemented in this study to evaluate the physical appearance of broilers in a poultry house. The system focuses on comb color, a key indicator of broiler health, as changes in comb color can be associated with various poultry diseases. To minimize pathogen cross-contamination, monitoring systems that avoid unnecessary interaction between poultry and farmers are crucial. The video surveillance system in this study includes meticulously color-calibrated cameras to ensure accurate color measurement, addressing the limitations of color image sensors and the challenges of monitoring dynamic objects like animals.[8]

The system employs the YOLOv4 deep learning algorithm to automatically extract comb colors from a large number of images captured by the surveillance system. Training data was collected from 4K cameras under various lighting and weather conditions. Manual cropping operations were performed to label chicken combs for training the YOLOv4 model. The model achieved a precision of 90% and a recall of 89% for chicken comb detection. The image data was stored for up to 90 days, enabling long-term analysis of comb color behavior during broiler growth. This system provides a technique for automatically extracting comb colors, which can assist professionals in making color-related broiler health diagnoses and improve poultry house management.[8]

2.3 REVIEW OF RELEVANT TECHNOLOGIES

In the field of early detection of disease in farming there are many technologies which are used which include the use of computer vision

2.3.1 Advanced Biosensors for Pathogen Detection

Biosensors represent a promising technology for rapid and sensitive pathogen detection in livestock. These devices utilize sensitive and selective recognition properties of biomolecules such as DNA receptors, glycans, aptamers, and antibodies [1]. For example, biosensors have been developed for the detection of avian influenza viruses, leveraging the specificity of hemagglutinin (HA) surface protein for sialic acid glycan residues[1]. While some biosensing systems are commercially available, ongoing research focuses on developing portable, miniaturized, and multitargeting devices for field application. Future advancements in biosensor technology are expected to incorporate nanotechnologies, advanced nucleic acid amplification methods, and next-generation sequencing analysis[2][3]

A key advantage of biosensors is their potential for on-site disease detection without requiring laboratory facilities. Paper-based biosensors, for example, have been developed as low-cost, portable, and easy-to-use solutions for field applications. Additionally, nucleic acid-based biosensors have shown high specificity and sensitivity, making them suitable for early disease surveillance. However, the main challenge of biosensor technology lies in sample preparation, potential false positives or negatives, and the need for further field validation before large-scale implementation in poultry farms.[1]

2.3.2 AI and Deep Learning for Poultry Disease Detection

2.3.2.1 Machine Learning in Poultry Disease Detection

Machine learning (ML) has played a transformative role in Precision Livestock Farming (PLF) by enabling data-driven decision-making for animal health management. Traditional livestock farming relies heavily on the experience of farmers, whereas PLF integrates Information and Communication Technologies (ICT) and ML models to enhance productivity, sustainability, and animal welfare[8]. In poultry farming, ML techniques analyze large datasets from sensors, images, and environmental conditions to predict disease outbreaks. Commonly used models include Decision Trees, Support Vector Machines (SVM), and Random Forests, which classify broiler health conditions based on structured data such as feed intake, weight variations, and temperature fluctuations [8]. These models have been successfully used to detect abnormalities in broiler behavior, ensuring early intervention and reducing mortality rates. However, ML models often require manual feature extraction, making them less effective for unstructured data like images and sounds [8]. As a result, deep learning models have emerged as a more advanced approach for detecting poultry diseases automatically.

2.3.2.2 Deep Learning Model Poultry in Disease Detection

You Only Look Once (YOLO)

You Only Look Once (YOLO) is a real-time object detection system known for its speed and accuracy in localizing objects within images, making it useful for identifying specific anatomical features or abnormal postures in poultry as compared to other model such as FOMO and MobileNet.[10]. YOLO can be used to automatically detect specific anatomical features, such as combs, wattles, or eyes, and to identify abnormal postures or behaviors[19]. The ability of YOLO to process images quickly makes it suitable for real-time monitoring of large poultry flocks, enabling early detection of potential health issues and facilitating prompt intervention. Recent advancements in YOLO versions YOLOv4, YOLOv5, YOLOv7 have further improved its performance and efficiency.[12]

However, YOLO models are often computationally intensive, requiring significant processing power and memory. This can pose challenges in resource-constrained environments, such as of small scale efamrmers where access to high-performance computing may be limited.

MobileNet Deep Learning Model

MobileNet architectures are designed for mobile and embedded vision applications, prioritizing efficiency and speed [11].They are lightweight CNNs suitable for deployment in environments with limited computational resources.Existing technologies utilize MobileNetSSD for real-time infection detection based on broiler movement analysis, achieving promising results[20]. However, MobileNet models are primarily focused on classification and may not be as effective as YOLO for precise object localization. Furthermore, while efficient, MobileNet models may still require more computational resources than FOMO, especially when deployed on very low-power microcontrollers

FOMO(Faster Objects, More Objects) Deep Learning Model

In contrast to YOLO and MobileNet, 'Faster Objects, More Objects' (FOMO) models are specifically designed for deployment on highly resource-constrained devices, such as microcontrollers and edge computing platforms making them suitable for deployment on embedded systems poultry farms[4]. FOMO prioritizes speed and efficiency through architectural optimizations, enabling real-time image classification with minimal computational overhead. . FOMO works by leveraging the TinyML optimized features for low latency and energy efficiency.FOMO is converted from the Tensorflow to lite version which is now used in microcontrollers for object classification or detection[4]

2.4 RESEARCH GAP

The preceding review has highlighted significant advancements in poultry disease detection, including the integration of sensor technologies, computer vision, and machine learning. These works have substantially improved the ability to monitor broiler health and detect diseases early. However, despite these advancements, several research gaps.

While current environmental monitoring systems effectively track real-time parameters like temperature and humidity [6], they lack predictive capabilities. This constitutes a significant gap, as environmental stressors often precede disease outbreaks. The ability to forecast potential stress events and implement timely interventions is crucial for preventing disease and minimizing economic losses.

Despite the advancements in wearable sensor technology for broiler monitoring, significant research gaps persist in optimizing sensor attachment methods to minimize impacts on bird welfare and behavior. Further studies are needed to explore novel attachment materials and techniques that improve retention without affecting physical integrity or data accuracy. Quantifying the effects of sensor attachment on locomotion, posture, social interactions, and natural behaviors is essential Additionally, research should focus on developing age-specific attachment solutions and mitigating potential data noise from loose sensor fittings.[5]

Current automated disease detection systems often focus on single modalities, such as sound analysis [13]. This approach overlooks the complex nature of poultry diseases, which typically manifest through a combination of symptoms. A critical research gap exists in the development of multimodal systems that integrate diverse data streams, including sound and image. Such integration would improve the accuracy and reliability of disease detection, enabling earlier and more effective interventions.[13][7].This research will develop a hardware system that captures broiler sounds and employs machine learning to detect respiratory disease indicators in real-time

While the MobileNetSSD model demonstrates promising results for real-time infection detection in broiler farms through movement analysis [11], it primarily focuses on movement data. A research gap exists in the integration of other potentially valuable data sources, such as audio analysis and image , which could provide a more comprehensive and accurate assessment of broiler health. By only focusing on movement, the current systems can miss other important indicators of disease.[13]This research will develop a system that integrates image analysis of droppings with sound analysis of broiler vocalizations, using machine learning to identify disease patterns

While advanced disease detection systems are being developed, there is a lack of focus on delivering actionable recommendations and alerts to small-scale farmers in a user-friendly manner. Existing systems often require technical expertise or access to sophisticated equipment. A research gap exists in developing accessible platforms that provide real-time alerts and actionable recommendations through dashboards and SMS, tailored to the needs of small-scale farmers. This research will develop a web dashboard and SMS alert system that provides farmers with real-time disease alerts and actionable recommendations based on the integrated analysis of droppings, images, and sounds

These identified research gaps highlight the need for an integrated and accessible system that is able to to detect contagious broiler diseases through the analysis of droppings, images, and sounds. By addressing these gaps, this research aims to empower small-scale farmers with the tools and information necessary to improve broiler health, reduce mortality rates, and enhance farm productivity.

2.5 RESEARCH THEORY

This section outlines the theoretical foundations underpinning the development of an early disease detection system for broilers. The system leverages machine learning and deep learning techniques to analyze broiler images, sound, and droppings for signs of coccidiosis and Newcastle disease. The core theoretical areas include pattern recognition, deep learning, signal processing, computer vision, and concepts from broiler health and veterinary science.

2.5.1 Pattern Recognition Theory

Pattern recognition theory provides the fundamental framework for this research. It focuses on the ability of systems to identify and classify regularities within data. In this project, deep learning algorithms maybe employed to recognize patterns indicative of disease in multimodal data streams.

Feature Extraction

A crucial aspect of pattern recognition is feature extraction, which involves transforming raw data into a set of meaningful features that deep learning models can utilize. For image data, Convolutional Neural Networks (CNNs), inherent in both YOLO and FOMO, automatically learn hierarchical features from images through convolutional layers, pooling layers, and activation functions [14]. For sound data, acoustic feature extraction techniques, such as Mel-Frequency Cepstral Coefficients (MFCCs), are employed to represent the spectral envelope of audio signals.[15]

Classification and Object Detection

This research may utilize both classification and object detection. FOMO models are employed for image classification, assigning a label to an image based on its overall content[4]. YOLO models, on the other hand, perform object detection, identifying and localizing specific signs of disease within an image . Both approaches fall under supervised learning, where models learn from labeled data.[10][14]

2.5.2 Deep Learning Theory

Deep learning theory forms a core component of this research.Deep learning models, particularly CNNs, have demonstrated exceptional capabilities in image and potentially sound analysis.

Convolutional Neural Networks (CNNs)

CNNs are a class of deep neural networks that excel at processing structured data like images. Their architecture, comprising convolutional layers, pooling layers, and activation functions, enables them to learn hierarchical representations of visual features. This hierarchical learning is crucial for identifying complex patterns associated with broiler diseases.[14]

YOLO (You Only Look Once)

YOLO is a real-time object detection system that excels at simultaneously predicting bounding boxes and class probabilities. Its architecture is designed for speed and efficiency, making it suitable for real-time monitoring. YOLO utilizes anchor boxes to predict objects of different sizes and shapes within an image [16][10].

FOMO (Faster Objects, More Objects)

FOMO models are optimized for resource-constrained devices, such as edge computing platforms. They prioritize speed and efficiency, making them suitable for deployment in poultry farm environments with limited computational resources. FOMO achieves this efficiency by making architectural changes that reduce the computational load [4].

Transfer Learning

Pre-trained models may be utilized in this research, leveraging the concept of transfer learning. Transfer learning involves adapting a model trained on a large, general dataset to a specific task, such as broiler disease detection. This approach can accelerate training and improve performance, especially when dealing with limited labeled data.[10]

2.5.3 Signal Processing Theory

Signal processing theory provides the framework for analyzing and interpreting sound data captured by the system. While deep learning models can learn features directly from raw audio, initial signal processing steps may be beneficial. Techniques such as Mel-Frequency Cepstral

Coefficients (MFCCs) are often used to extract perceptually relevant features from audio signals, representing the spectral envelope of the sound [15].

2.5.4 Computer Vision Theory

Computer vision theory underpins the analysis and interpretation of image data within this research. Image preprocessing techniques may be applied to enhance image quality and prepare images for analysis. These techniques may include noise reduction and image enhancement methods.

2.5.5 Broiler Health and Veterinary Science Concepts

This research is grounded in the understanding of broiler health and veterinary science, specifically focusing on coccidiosis and Newcastle disease.

Coccidiosis is a parasitic disease affecting the intestinal tract of broilers, characterized by symptoms such as diarrhea, reduced growth rate, and increased mortality. Visual signs may include bloody droppings, ruffled feathers, and lethargy [17][18].

Newcastle disease is a highly contagious viral disease affecting poultry, characterized by respiratory, digestive, and nervous system symptoms. Visual signs may include coughing, sneezing, nasal discharge, and paralysis .

2.6 CONCLUSION

This section has examined the current landscape of poultry disease detection, highlighting the evolution from traditional methods to advanced sensor-based and AI-driven systems. While significant progress has been made in areas such as environmental monitoring, wearable sensors, and computer vision, several key research gaps remain. Notably, the need for multimodal systems that integrate diverse data streams and the development of predictive models for proactive intervention are critical. Furthermore, the challenges faced by small-scale farmers in, Zimbabwe, necessitate robust and cost-effective solutions.

This research aims to address these gaps by developing an early disease detection system that leverages FOMO and YOLO deep learning models for multimodal analysis of broiler sounds, images, and droppings. By integrating these technologies, the proposed system will provide real-time alerts and actionable recommendations, empowering farmers to make informed decisions and improve broiler health. The theoretical foundations of this research, including pattern recognition, deep learning, signal processing, computer vision, and veterinary science, provide a solid framework for the development of a reliable and effective disease detection system. Ultimately, this research seeks to contribute to the sustainability and productivity of small-scale broiler farming in Zimbabwe and beyond.

. References:

1) Vidic, J., Manzano, M., Chang, CM. et al. Advanced biosensors for detection of pathogens related to livestock and poultry. Vet Res 48, 11 (2017). https://doi.org/10.1186/s13567-017-0418-5
2) Vidic, J., Manzano, M., Chang, C. M., & Jaffrezic-Renault, N. (2017). Advanced biosensors for detection of pathogens related to livestock and poultry. Veterinary research, 48(1), 11. https://doi.org/10.1186/s13567-017-0418-5
3) Recent Advancement in Biosensors Technology for Animal and Livestock Health ManagementSuresh Neethirajan, Sheng-Tung Huang, Satish K. Tuteja, David KeltonbioRxiv 128504; doi: https://doi.org/10.1101/128504
4) Dharani, A., Kumar, S.A. & Patil, P.N. Object Detection at Edge Using TinyML Models. SN COMPUT. SCI. 5, 11 (2024). https://doi.org/10.1007/s42979-023-02304-z
5) Khan, I., Peralta, D., Fontaine, J., Soster de Carvalho, P., Martos Martinez-Caja, A., Antonissen, G., Tuyttens, F., & De Poorter, E. (2025). Monitoring Welfare of Individual Broiler Chickens Using Ultra-Wideband and Inertial Measurement Unit Wearables. *Sensors*, *25*(3), 811. https://doi.org/10.3390/s25030811
6) Abo-Al-Ela, H. G., El-Kassas, S., El-Naggar, K., Abdo, S. E., Jahejo, A. R., & Al Wakeel, R. A. (2021). Stress and immunity in poultry: light management and nanotechnology as effective immune enhancers to fight stress. Cell stress & chaperones, 26(3), 457–472. https://doi.org/10.1007/s12192-021-01204-6
7) APA StyleGarrido, L. F. C., Sato, S. T. M., Costa, L. B., & Daros, R. R. (2023). Can We Reliably Detect Respiratory Diseases through Precision Farming? A Systematic Review. *Animals*, *13*(7), 1273. https://doi.org/10.3390/ani13071273

8) Comb Color Analysis of Broilers Through the Video Surveillance System of a Poultry House https://doi.org/10.1590/1806-9061-2023-1891

9) García, R., Aguilar, J., Toro, M., Pinto, A., & Rodríguez, P. (2020). A systematic literature review on the use of machine learning in precision livestock farming. Computers and Electronics in Agriculture, 179, 105826.

10) S. . Tamang, B. . Sen, A. . Pradhan, K. . Sharma, and V. K. . Singh, “Enhancing COVID-19 Safety: Exploring YOLOv8 Object Detection for Accurate Face Mask Classification”, Int J Intell Syst Appl Eng, vol. 11, no. 2, pp. 892–897, Feb. 2023.

11) Doko, L. A., Jiya, E. A., & Olanrewaju, O. M. (2024). Real-Time Infection Detection System in Broiler Farm using MobileNetSSD Model. Journal of Science Research and Reviews, 1(1), 78-86.

12) Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., ... & Adam, H. (2017). Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861*

13) Alqudaihi, Kawther S et al. “Cough Sound Detection and Diagnosis Using Artificial Intelligence Techniques: Challenges and Opportunities.” IEEE access : practical innovations, open solutions vol. 9 102327-102344. 15 Jul. 2021, doi:10.1109/ACCESS.2021.3097559.

14) Bhatt, D., Patel, C., Talsania, H., Patel, J., Vaghela, R., Pandya, S., Modi, K., & Ghayvat, H. (2021). CNN Variants for Computer Vision: History, Architecture, Application, Challenges and Future Scope. *Electronics*, *10*(20), 2470. https://doi.org/10.3390/electronics10202470

15) Cuan, K., Zhang, T., Li, Z., Huang, J., Ding, Y., & Fang, C. (2022). Automatic Newcastle disease detection using sound technology and deep learning method. Computers and Electronics in Agriculture, 194, 106740.

16) Vijayakumar, A., Vairavasundaram, S. YOLO-based Object Detection Models: A Review and its Applications. Multimed Tools Appl 83, 83535–83574 (2024). https://doi.org/10.1007/s11042-024-18872-y

17) Blake, D.P., Knox, J., Dehaeck, B. et al. Re-calculating the cost of coccidiosis in chickens. Vet Res 51, 115 (2020). https://doi.org/10.1186/s13567-020-00837-2

18) To control Coccidiosis in broiler chickens in fowrun https://www.wur.nl/en/research-results/research-institutes/livestock-research/show-wlr/to-control-coccidiosis-in-broiler-chickens.htm#:~:text=Coccidiosis%20is%20an%20important%20health,be%20alternated%20by%20using%20vaccination.

19) Elmessery, W. M., Gutiérrez, J., Abd El-Wahhab, G. G., Elkhaiat, I. A., El-Soaly, I. S., Alhag, S. K., Al-Shuraym, L. A., Akela, M. A., Moghanm, F. S., & Abdelshafie, M. F. (2023). YOLO-Based Model for Automatic Detection of Broiler Pathological Phenomena through Visual and Thermal Images in Intensive Poultry Houses. *Agriculture*, *13*(8), 1527. https://doi.org/10.3390/agriculture13081527

20) L. A. Doko, E. A. Jiya, and O. M. Olanrewaju, “Real-Time Infection Detection System in Broiler Farm using MobileNetSSD Model”, JOSRAR, vol. 1, no. 1, pp. 78–86, Dec. 2024, doi: 10.70882/josrar.2024.v1i1.21.